vnv_coverage_completeness

HelixCode V&V Coverage-Completeness Map (§11.4.118 honesty layer)

Property	Value
Date	2026-06-24
Server	`http://localhost:18080` (fresh, autoboot PG/Redis/Ollama)
Method	READ-ONLY: read `docs/qa/helixcode_vnv_execution_plan.md`, the two Makefiles, `helix_code/qa-results/*` cycle logs, `docs/features/Status.md`
Author	coverage-completeness subagent, data-only (§11.4.6 / §11.4.3)
Purpose	Make this cycle's V&V boundary EXPLICIT. "No other issues" is only credible with this enumerated set. NOT a claim of completeness.

This map enumerates what ran, what was honestly SKIP-justified, and what was left un-attempted (SHOULD-RUN). It cross-references the executed `qa-results` logs against the full test-type catalogue in the execution plan.

---

0. Honesty correction vs the cycle brief (§11.4.6 — data over narrative)

The originating brief described "unit wave (133 ok)" and "constitution sweep 16/16". The captured logs say otherwise — recorded here so the cycle is not misrepresented as cleaner than it is:

- **Unit wave (`helixcode_unit_wave_20260624_110425.log`):** 133 packages `ok`, **but the overall run FAILED** — `dev.helix.code/internal/cognee` FAILs (60.4s), and 5 hardware GPU-probe unit tests FAIL on this host: `TestGetGPUUsage_ProbeChain_NvidiaPreferred`, `…_FallsBackToAMD`,

`TestProbeAppleGPU_ParsesIoregOutput`, …`_HandlesMultipleGPUs`, …`_HandlesWhitespaceVariants`. The GPU-probe failures are host-hardware dependent (no nvidia-smi/rocm; Apple ioreg variant parsing) → should be `uname -s`-gated SKIPs per §11.4.81, not hard FAILs. `cognee` FAIL needs a §11.4.102 root-cause (it also recurs in the stress/chaos `cognee_*` dirs).
- **Constitution sweep (`constitution-sweep.log`): 13 PASS / 3 FAIL**, NOT 16/16. Failing gates are all doc-sync/governance, not product: **G7** (feature commits lack `docs/qa/<run-id>/` evidence), **G12** (Issues/Fixed summary docs stale), **G14** (`docs_chain verify --all` drift). G16 (challenge matrix) PASS.
- **e2e (phase3) PASS** confirmed (`e2e_20260624_112208.log`, EXIT 0).
- **e2e-challenges FAILED** (`e2e_challenges_…/runner.log`): 0.00% success, 4 failed + 2 validation-failed, **Total Tokens: 0** — the 0-token figure means no real LLM generation tokens flowed through the challenge path (timeout bug under fix; treat as a live defect, not a pass).
- **HelixQA api: 169 PASS / 25 SKIP / 18 FAIL** confirmed. NOTE: the `helixqa_findings_triage.md` triaged only **7** of the 18 FAILs (2 bank-drift + 5 `/server/info` exposure gaps). The remaining FAILs include **auth-login 401 "invalid credentials"** (e.g. HXC-WRK-010) — a server seed/credential mismatch, NOT yet triaged; un-triaged FAILs are an open coverage gap inside the bank run.

---

1. RAN this cycle (verdict captured)

Test type	Command	Verdict (from log)	Evidence
Compile / `go vet`	`make verify-compile` / vet	PASS	`go-vet.log`, `helixcode_govet_20260624_110433.log`
Unit (hermetic)	`make test`	**133 ok but FAIL** (cognee + 5 GPU-probe)	`helixcode_unit_wave_20260624_110425.log`
Verifier unit	`make test-verifier-unit`	ran (PASS)	`verifier-unit.log`
Integration	`make test-integration-full`	PASS (incl. IDOR green)	`integration_*_20260624_11*.log`
E2E (phase3)	`make test-e2e-full` (phase3)	PASS	`e2e_20260624_112208.log`
E2E challenges	`tests/e2e/challenges -all`	**FAIL** (0%, 0 tokens, timeout bug)	`e2e_challenges_20260624_112611/runner.log`
HelixQA api banks	`helixqa http --banks … api…`	169 PASS / 25 SKIP / 18 FAIL	`helixqa_api_20260624_112930/helixqa.log`
Constitution sweep	governance sweep	**13/16 PASS** (G7/G12/G14 FAIL)	`constitution-sweep.log`

Test type	Command	Verdict (from log)	Evidence
Benchmark sanity	`make test-benchmark` (sanity)	ran	`bench-sanity.log`, `bench-sanity-challenges.log`
Challenges anti-bluff	scan + anchors	ran	`challenges-anti-bluff-scan.log`, `…-anchors.log`
Stress + chaos	`stress-chaos` (partial)	**PARTIAL** — ~1500 scenario evidence dirs present (`client_http_*`, `cognee_*` etc.) but no captured single suite verdict; `cognee` chaos overlaps the unit FAIL	`qa-results/20260624T09*Z/` dirs

---

2. SKIP-justified (§11.4.3 — honest, missing dep/SDK/credential named)

Item	Why SKIP-justified	Un-SKIP needs
Real cloud-LLM generation (OpenAI/Anthropic/Gemini/Groq/Mistral/Cohere/Azure/Bedrock)	`.env.full-test` ships **mock** keys → mock-llm-server; no real wire calls	real API keys in non-mock `.env`
Verifier integration (`make test-verifier-integration`)	LLMsVerifier service @ `:8081` not booted in this run	boot verifier service (own submodule — borderline-cheap, see §4)
HelixQA mobile/TV banks (`full-qa-android*`, `nexus-mobile-ios`, `capture-android`)	no Android/iOS device/emulator/simulator + ADB/Xcode; iOS has no Xcode project	device/emulator + platform SDK
HelixQA desktop banks (`nexus-desktop-macos/windows/linux`)	desktop GUI build + host display required	per-OS desktop binary on a display
Mobile client tests (Android/Aurora HAP/Harmony HAP)	device/emulator required	device/emulator
Desktop (Fyne) GUI runtime tests	host display server required (CI headless)	display (`desktop-nogui` build is partial proxy)
		real Xiaomi key

Item	Why SKIP-justified	Un-SKIP needs
Xiaomi provider challenge (`xiaomi_provider_challenge.go`, `qa-integration/xiaomi_test_bank.yaml`)	real Xiaomi provider credential	
Security scanners snyk / sonarqube	need `SNYK_TOKEN` / `SONAR_TOKEN`	tokens
Sibling-project HelixQA banks (`atmosphere*`, `boba-*`, `openclawing2-*`, `yole-*`, `nexus-*`)	not HelixCode — different SUT	out of scope

---

3. SHOULD-RUN — un-attempted, NOT a justified SKIP (un-exercised but runnable)

These had no missing dependency this cycle; they were simply not run.

Item	Command	Why it should have run	Cost vs current infra
Security suite	`make test-security-full`	core security test type, infra present	CHEAP (server-touching, serial on :18080)
Stress + chaos (complete)	`make stress-chaos` + `make stress-chaos-infra`	only partial evidence dirs exist, no captured suite verdict	CHEAP (unit hermetic; infra needs PG+Redis, present)
Load / perf	`make test-load-full`	perf/load test type entirely absent	CHEAP (server-touching, serial)
Coverage (-race)	`make test-coverage`	no coverage profile captured this cycle	CHEAP (hermetic)
Static security scanners	`make scan-gosec scan-trivy scan-grype` (root)	open-source, no creds	CHEAP (no infra)
Remaining HelixQA api banks	helixqa http --banks <not-yet-run>	only one api invocation captured; mcp-fs / persistent-memory / protected-subresources / static-routes / cors banks not confirmed run	CHEAP (serial on :18080)

Item	Command	Why it should have run	Cost vs current infra
HelixQA web banks	`helixqa run --platform web --browser-url …`	web client browser flows have ZERO V&V this cycle	BORDERLINE (needs selenium :4444 / chromedp :9222 up + browser-url reaching the server)
Challenges submodule full	`cd submodules/challenges && make qa-all` (+ `new-capabilities`)	only anti-bluff scan + bench-sanity ran	CHEAP for `qa-all` (hermetic); capability runs need :8080
Unit FAIL remediation	fix/SKIP-gate the 5 GPU-probe tests (§11.4.81) + root-cause `cognee`	unit wave is not green	CHEAP (unit-level)

---

4. Unknown-unknown surface — subsystems with NO V&V evidence this cycle

Cross-referencing `docs/features/Status.md` Areas (257 service rows, 114 infrastructure, 134 submodule, 30 cli / 7 tui / 4 web / 6 desktop / 8 mobile) against what actually ran, these had NO runtime V&V this cycle (at most hermetic unit coverage, which the §11.4.108 four-layer rule does not count as runtime-on-clean-target evidence):

1. **Web client (browser flows)** — api-only HelixQA ran; no selenium/chromedp browser session drove the web UI.
2. **Desktop (Fyne) GUI runtime** — no run (display gap).
3. **Mobile clients (Android/iOS/Aurora/Harmony)** — no run (device/SDK gap).
4. **LSP / skills / ensemble capabilities** — `internal/lsp`, `internal/skills`, `internal/ensemble` packages are ABSENT (per triage); banks assert their `*_enabled` flags → these are real gaps with NO working V&V, only FAILing assertions. `plugins` + streaming exist but are not advertised in `/server/info` (info-exposure gaps).
5. **~65 owned submodules** — of the 134 submodule rows, only `helix_qa` (used as the framework) and `challenges` (anti-bluff scan) were exercised. The rest (security, helix_agent, helixspecifier, panoptic, containers, docs_chain, llms_verifier runtime, etc.) had no V&V dispatched in this cycle.
6. **cli_agents ported capabilities** — Status.md lists a “Ported cli_agents” slice; no dedicated port-capability run this cycle.
7. **Many internal services covered only by hermetic unit tests** (no real-infra integration/e2e this cycle): e.g. `deployment`, `discovery`, `hardware` (GPU-probe FAILing), `focus`, `editor`, `repomap`, `template`, `rules`, `hooks`, `version`. Their “working” status rests on unit tests, not runtime evidence.
8. **Real provider matrix** — only Ollama (local) + mock-llm-server exercised; the CONST-039 “all providers” surface (OpenAI/Anthropic/Gemini/DeepSeek/Groq/ Mistral/xAI/OpenRouter/Llama.cpp) has no real-wire V&V (mock keys).

Honest boundary (§11.4.6): this list reduces the unknown-unknown surface; it does NOT prove the un-exercised subsystems work. They are honest coverage gaps, not implied-clean.

---

5. Prioritized "cheaply-runnable-now" candidates (additional V&V wave)

Against the CURRENT booted infra (`:18080` + PG/Redis/Ollama, no new deps), serialized on the single server per §11.4.119:

1. **`make test-security-full`** — highest value, server-touching, serial.
2. **`make stress-chaos` then `make stress-chaos-infra`** — complete the partial run + capture a single suite verdict (and re-confirm the `cognee` chaos vs unit FAIL relationship — likely one root cause, §11.4.102).
3. **`make test-load-full` + `make test-benchmark`** — fills the entirely-absent perf/load type; load is serial, benchmark hermetic [bg].
4. **Remaining HelixQA api banks** in ONE `helixqa http` invocation (the api banks not yet confirmed run) — serial on :18080.
5. **`cd submodules/challenges && make qa-all`** — hermetic, parallel-safe [bg].
6. **Root static scanners `make scan-gosec scan-trivy scan-grype`** — no infra, parallel-safe [bg].
7. **`make test-coverage`** — hermetic [bg].
8. **Unit-level fixes:** `uname -s`-gate the 5 GPU-probe tests (§11.4.81) and root-cause the `cognee` unit FAIL — cheap, makes the unit wave green.

Genuinely blocked (stay §11.4.3 SKIP this wave): real cloud-LLM generation (no keys), mobile/desktop client tests (devices/SDK/display), Xiaomi (real key), snyk/sonarqube (tokens). **Borderline (cheap only if their service/containers are booted):** verifier-integration (`:8081`, own submodule — boot it to un-SKIP) and HelixQA web banks (selenium :4444 / chromedp :9222 + browser-url reaching the server on the host-remapped :18080).